

CYBERMIND Architecture Document

Smart India Hackathon 2026 | Problem Statement SIH26153 | github.com/IconoclasticDev/CYBERMIND

1. System Overview

CYBERMIND is a four-tier, local-first platform that models a protected network as a sequence of host-communication graphs. A graph-temporal world model forecasts short-horizon attack risk, Parallel Futures compares defensive interventions in simulation, and the Strix Lab validates attack hypotheses and re-tests approved defences in an isolated sandbox. All core processing runs locally with no cloud model endpoint. The diagram shows the four tiers, the main left-to-right data path and the two feedback loops.

Observe > Forecast > Simulate > Validate > Defend > Re-test > Report

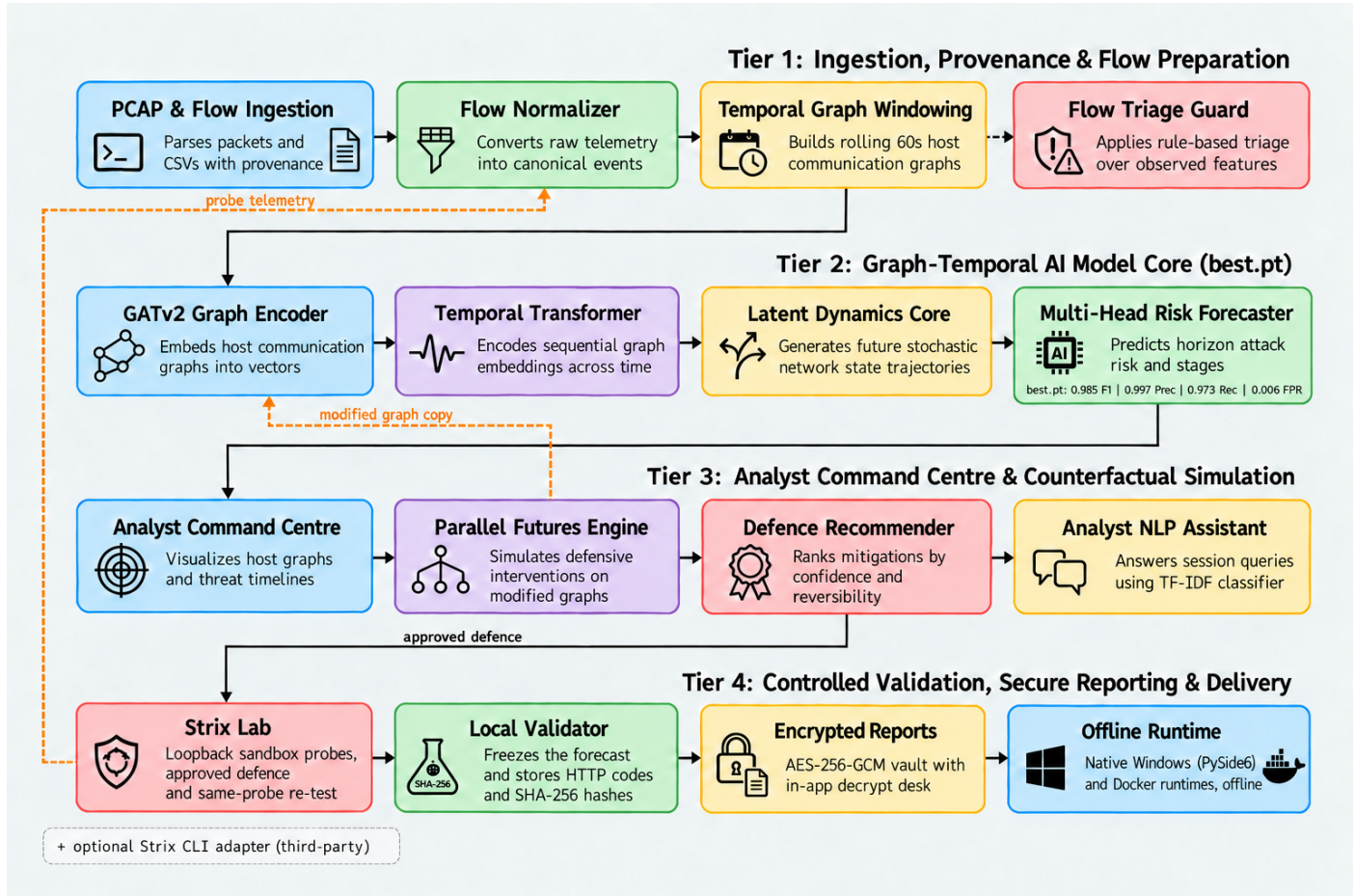


Figure 1. CYBERMIND four-tier architecture. Solid arrows: data flow. Dashed black: side branch. Dashed orange: feedback loops (probe telemetry re-enters at the Flow Normalizer, modified graph copies re-enter at the GATv2 encoder).

2. Architecture Layers

Tier	Components	Responsibility
1 Ingestion	Ingestion service, normalizer, temporal graph windowing, flow triage guard	Records provenance, builds canonical events and rolling host-communication graphs, and keeps rule-based flow triage separate from model forecasts.
2 Model core	GATv2 encoder, temporal Transformer, latent dynamics, risk and stage heads	Encodes recent graph history and produces stochastic future trajectories, window-level risk, coarse stage estimates and uncertainty proxies.
3 Analyst layer	Command Centre, Parallel Futures, defence recommender, offline assistant	Links evidence to forecasts, compares no-action and intervention rollouts, ranks reversible defences and answers bounded questions about the loaded case.
4 Validation and delivery	Strix Lab, local validator, AES-256-GCM vault, desktop and Docker runtimes	Tests attack hypotheses and defences in a sandbox, records tamper-evident evidence, encrypts incident briefs and keeps processing local.

3. End-to-End Flow

1	Analyst imports a PCAP or flow CSV	9	Sandbox responses show whether the vector was accepted or blocked
2	Backend records provenance and builds canonical flow events	10	Probe telemetry updates the case and the model risk view
3	Events form overlapping host-communication graph windows	11	A confirmed vulnerability produces a proposed remediation
4	best.pt forecasts future graph-window risk and coarse stage	12	Analyst approves the lab defence
5	Analyst inspects topology, flows, timeline and evidence	13	The same probe is repeated to verify the mitigation
6	Parallel Futures compares no-action and modified-graph rollouts	14	Forecast, evidence, decisions and validation enter the incident brief
7	Defence recommender ranks reversible interventions	15	The brief is encrypted locally with AES-256-GCM
8	Analyst optionally runs a controlled Strix Lab probe	16	Recipient imports and decrypts it in CYBERMIND with the separate key

4. Component Detail

Ingestion and graph construction

The analyst selects a .pcap, .pcapng, .cap or flow .csv file; nothing is preloaded. The FastAPI ingestion service records provenance (uploaded observation, dataset replay, synthetic scenario, controlled lab probe, validation record, approved action), extracts packets or matches flow columns, and builds a canonical event: endpoints, ports, protocol, timestamps, duration, packet and byte counts and optional source labels. Missing CSV timestamps use a disclosed assumed interval, and raw captures stay Unassessed unless valid labels exist. Events form directed host-communication graphs in 60 s windows with a 30 s stride (up to 64 retained, at least 3 required to forecast). Features include volumes, rates, protocol and port information, duration, peer and port diversity, inter-arrival statistics and scanning behaviour. Input lacking fields the checkpoint requires is rejected, never filled with fabricated values.

Graph-temporal world model (best.pt)

Stage	Processing	Output
Graph encoder	GATv2 attention over each host-communication graph	One embedding per window
Temporal encoder	Transformer over the ordered graph embeddings	Latent network state
Dynamics rollout	Learned diagonal-Gaussian transition applied recursively over multiple samples	Stochastic future states, dispersion
Prediction heads	Binary infiltration-risk, coarse-stage and future-state heads; optional gradient-based sensitivity evidence	Horizon risk, stage distribution, OOD proxy

If the checkpoint is missing or invalid, forecasting is reported unavailable; no substitute scores are generated.

Analyst layer and counterfactual simulation

The React Command Centre reads events, graphs, forecasts and validation records from the local FastAPI backend over REST and WebSocket. Parallel Futures copies the current graph, applies an in-memory intervention (host isolation, port blocking, edge restriction or rate limiting) and reruns the same model against a no-action baseline. The recommender ranks options by predicted risk change, confidence proxy, reversibility and a collateral-impact heuristic. Outputs are model-based comparisons for analyst review, not measured causal effects or executed firewall rules. The offline assistant (TF-IDF character n-grams, scikit-learn intent classifier) answers read-only questions about the loaded case.

Strix Lab and local validation

The built-in Strix Lab sends bounded, non-destructive HTTP probes only to an isolated loopback sandbox. Each verdict comes from the sandbox's real response (Succeeded, Blocked, or Unreachable, where no verdict is asserted) and becomes a telemetry event that updates the forecast. When a vulnerability is confirmed, a proposed remediation is reviewed and approved by the analyst, applied in the lab, and the same probe is re-run; a Blocked result is evidence that the mitigation changed the sandbox's behaviour. A local validation engine also freezes each forecast, calls four bounded endpoints on the bundled sandbox and stores HTTP status codes and SHA-256 response hashes with the forecast ID and model version. An optional adapter for the third-party Strix CLI supports registered, authorized targets behind a safety gate; it needs Docker and an LLM provider and is not part of the offline core.

SSH brute force T1110.001 CWE-307	SQL injection probe T1190 CWE-89	SMB lateral access T1021.002 CWE-285	C2 beaconing T1071.001 CWE-200
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Protected reporting and deployment

Incident briefs are encrypted locally with AES-256-GCM using a fresh random 256-bit key and 96-bit nonce from Web Crypto. The JSON envelope holds the nonce and ciphertext, never the key; the recipient imports it and enters the separately shared key, and a wrong key or altered ciphertext fails decryption. CYBERMIND ships as a Docker-backed Windows launcher (canonical release, requires Docker Desktop), a native Windows build (PyInstaller, PySide6/Qt WebEngine, no Docker) and prebuilt Linux x86-64 images run with Docker Compose. Core forecasting, Strix Lab and reporting need no cloud endpoint.

5. Evaluation

Capture-grouped CIC-IDS2018 protocol: four observed windows predict four unseen future windows, complete capture groups are separated across training, validation and test, and the threshold is fixed at 0.5.

Held-out binary	F1	Precision	Recall	False-positive rate
CYBERMIND best.pt	0.9849	0.9970	0.9730	0.0057
Feature-matched logistic baseline	0.9266	0.9314	0.9219	0.1314

Coarse-stage macro-F1 is 0.2443, well below binary-risk performance, so stage output is a research estimate. Evaluation so far uses CIC-IDS2018 only; zero-shot tests on CTU-13 and UNSW-NB15 are planned.

6. Operational Boundaries

- Forecasts apply to aggregate graph windows, not verified malicious hosts or flows; flow-triage bands are rule-based, not best.pt output.
- Rollout variance is an uncertainty proxy, not a calibrated confidence, and Parallel Futures is a model simulation, not proof of causal effect.
- Probes are bounded, non-destructive and limited to the loopback sandbox unless an authorized target is registered. A sandbox re-test shows the mitigation worked there, not that it is safe in production, and sandbox activity never retrains best.pt.
- Consequential defensive actions remain under analyst control. Production use needs security review, wider external validation, threshold calibration, key-exchange procedures and reviewed model updates.